Autoencoder Asset Pricing: Out-of-Publication-Sample Evidence and Extensions

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Abstract

We replicate and extend the conditional autoencoder asset pricing model of Gu et al. (2021). Using 91 firm-level characteristics and monthly returns from CRSP for the period 1972–2023, we train conditional autoencoders of varying depth and latent dimension and evaluate them out of sample from 2011–2023. Our best model (CA_2, $K = 6$) achieves a total R^2 of 25.1% and a predictive R^2 of 0.65%, with a predictive long-short Sharpe ratio of 2.13 that exceeds a characteristics-only benchmark of 1.24. We extend the original framework in two directions. First, we test whether the model's reconstruction error constitutes a tradable anomaly signal: portfolio sorts reveal a monotonic decline in returns from low- to high-anomaly quintiles, but the spread does not appear statistically significant. Second, we propose energy-based and factor-disentanglement regularizers and find that modest penalties can improve predictive R^2 without degrading total fit. We also conduct a backtest of the predictive long-short strategy, documenting cumulative returns, drawdowns, factor alphas, and portfolio turnover.

1. Introduction

Traditional asset pricing models represent expected returns as linear combinations of exposures to latent risk factors. The Capital Asset Pricing Model (CAPM) posits a single market factor, while the Fama-French extensions add size and value portfolios. One limitation of these static models is that factor loadings are assumed constant, even though a firm's risk profile evolves with its characteristics.

Instrumented principal component analysis (IPCA), introduced by Kelly et al. (2019), addresses this by allowing betas to vary as linear functions of observable firm characteristics. Gu et al. (2021) extend this with a conditional autoencoder that replaces linear betas with a deep neural network that models nonlinearities. Their architecture separates the model into a beta network (mapping characteristics to factor loadings) and a factor network (projecting managed portfolio returns onto a low-dimensional factor space), with the predicted return formed as the inner product $\hat{r}_{i,t} = \hat{\boldsymbol{\beta}}(\mathbf{z}_{i,t-1}) \cdot \hat{\mathbf{f}}_t$. They show that this model dominates linear specifications across total R^2, predictive R^2, and long-short portfolio Sharpe ratios.

We make three contributions. First, we replicate the Gu, Kelly, and Xiu framework on an updated dataset extending through 2023, providing an out-of-publication-sample test of whether the conditional autoencoder's

advantages persist in a newer market environment. Second, we investigate whether the model's reconstruction error (the residual $\varepsilon_{i,t} = r_{i,t} - \hat{r}_{i,t}$) carries information about future returns, testing the hypothesis that large pricing errors signal regime transitions or temporary mispricing that can be exploited in a trading strategy. Third, we propose two auxiliary loss terms, a Sharpe penalty and a factor correlation regularizer, and investigate their effect on model performance through an ablation study.

Our results confirm that deep conditional autoencoders (CA_2 and CA_3) substantially outperform IPCA and shallow networks in explained return variation, with total R^2 reaching 25% compared to 6% for IPCA. However, the gap between contemporaneous and predictive Sharpe ratios is large (10.2 vs. 2.1 for our best model), indicating that the autoencoder's primary value is coming from cross-sectional stock selection through learned characteristic weightings rather than dynamic factor timing. The reconstruction error produces a monotonic return pattern across anomaly quintiles, but is not a statistically significant predictor of future returns, suggesting that the model's residuals reflect idiosyncratic noise rather than systematic mispricing. The codebase can be found here: `https://github.com/liamos7/fin580-autoencoders`.

2. Methodology

2.1 Model Design

The conditional factor model specifies excess returns as

$$r_{i,t} - r_{f,t} = \boldsymbol{\beta}(\mathbf{z}_{i,t-1}) \cdot \mathbf{f}_t + \varepsilon_{i,t}, \tag{1}$$

where $\mathbf{z}_{i,t-1} \in \mathbb{R}^P$ is a vector of lagged firm characteristics, $\boldsymbol{\beta} : \mathbb{R}^P \to \mathbb{R}^K$ maps characteristics to factor loadings, and $\mathbf{f}_t \in \mathbb{R}^K$ is the vector of latent factors. The conditional autoencoder of Gu et al. (2021) estimates $\boldsymbol{\beta}$ and $\mathbf{f}_t$ simultaneously by embedding a neural network in the beta specification.

The architecture divides into two components. The beta network passes P firm-level characteristics through s dense layers with batch normalization and ReLU activations, outputting K factor loadings per stock. We denote models with $s = 0, 1, 2, 3$ hidden layers as CA_0 through CA_3, with CA_0 corresponding to a linear beta specification, and IPCA serving as the fully linear baseline.

The factor network first reduces the N-dimensional return vector to $P + 1$ characteristic-managed portfolio returns via the mapping

$$\mathbf{x}_t = \left(\mathbf{Z}_{t-1}^\top \mathbf{Z}_{t-1}\right)^{-1} \mathbf{Z}_{t-1}^\top \mathbf{r}_t, \tag{2}$$

where $\mathbf{Z}_{t-1} \in \mathbb{R}^{N \times P}$ is the lagged characteristic matrix augmented with a constant column for the equal-weighted market portfolio. A single linear layer then projects $\mathbf{x}_t$ onto the K-dimensional factor space ($\hat{\mathbf{f}}_t = \mathbf{W}_f \mathbf{x}_t$). The linearity of the factor network is economically motivated by the idea that risk factors should be investable portfolios, and hence linear combinations of asset returns.

The predicted return is the inner product $\hat{r}_{i,t} = \hat{\boldsymbol{\beta}}_i \cdot \hat{\mathbf{f}}_t$, and the model is trained by minimizing mean

squared error with ℓ_1 regularization:

$$\mathcal{L} = \frac{1}{NT} \sum_{i,t} (r_{i,t} - \hat{r}_{i,t})^2 + \lambda_{\ell_1} \|\boldsymbol{\theta}\|_1. \tag{3}$$

.6.6

Figure 1: Conditional autoencoder architecture. The beta network (left) maps P firm characteristics through s dense layers to produce K factor loadings. The factor network (right) projects $P + 1$ managed portfolio returns onto K latent factors via a linear layer. The predicted return is the inner product of the two outputs.

2.2 Data

We obtain monthly individual stock returns from the Center for Research in Security Prices (CRSP) and merge them with a panel of 91 firm-level characteristics sourced from CRSP and the Chen and Zimmermann (2022) open-source cross-sectional asset pricing dataset. The full sample spans January 1972 to December 2023 and contains 4,221,335 stock-month observations. Following Gu et al. (2021), characteristics are cross-sectionally rank-normalized to $[-1, 1]$ each month. We split the sample into training (1972–2005), validation (2006–2010), and test (2011–2023) periods. The test period extends six years beyond the original paper's endpoint of 2016.

2.3 Evaluation Metrics

We evaluate models using three metrics from Gu et al. (2021). Total R^2 measures the fraction of return variation explained by contemporaneous factor realizations:

$$R^2_{\text{total}} = 1 - \frac{\sum_{i,t}(r_{i,t} - \hat{r}_{i,t})^2}{\sum_{i,t} r_{i,t}^2}. \tag{4}$$

Predictive R^2 replaces the contemporaneous factor realization $\hat{\mathbf{f}}_t$ with the prevailing mean of past factor estimates $\hat{\boldsymbol{\lambda}}_{t-1} = \frac{1}{t-1} \sum_{s=1}^{t-1} \hat{\mathbf{f}}_s$, isolating the return variation attributable to time-varying betas:

$$R^2_{\text{pred}} = 1 - \frac{\sum_{i,t}(r_{i,t} - \hat{\boldsymbol{\beta}}_i \cdot \hat{\boldsymbol{\lambda}}_{t-1})^2}{\sum_{i,t} r_{i,t}^2}. \tag{5}$$

Predictive Sharpe ratio sorts stocks into deciles each month by predicted return $\hat{\boldsymbol{\beta}}_i \cdot \hat{\boldsymbol{\lambda}}_{t-1}$ and computes the annualized Sharpe ratio of the long-short spread between the top and bottom deciles. We use a 60-month rolling window for $\hat{\boldsymbol{\lambda}}_{t-1}$ (seeded with training period factor estimates in the beginning of the test period).

As a benchmark, we also compute a characteristics-only Sharpe ratio that sorts stocks by the cross-sectional mean of all 91 rank-normalized characteristics without any autoencoder. The difference between the predictive and characteristics-only Sharpe ratios hence isolates the incremental value of the autoencoder's nonlinear characteristic weighting.

2.4 Training

Models are trained on Princeton's Adroit GPU cluster using the Adam optimizer with learning rate 10^{-3} and early stopping with a patience of 10 epochs. To address the non-convexity of the loss landscape, we train an ensemble of 10 models from different random initializations and average their predictions at inference time. We vary the number of latent factors $K \in \{1, 2, 3, 4, 5, 6\}$ and the network depth $s \in \{0, 1, 2, 3\}$, yielding 30 architecture-factor configurations.

2.5 Anomaly Detection Extensions

If the autoencoder's residual $\varepsilon_{i,t} = r_{i,t} - \hat{r}_{i,t}$ contains information about future returns, then stocks with large absolute residuals should earn abnormal returns in the following month. Each month, we sort stocks into quintiles by $|\varepsilon_{i,t}|$ and examine the return of each quintile at $t + 1$. Stocks are tracked by CRSP permno across months to ensure correct matching when the panel composition changes. We test the high-minus-low quintile spread using Newey-West standard errors with 3 lags, a standard t-test, and a Wilcoxon signed-rank test.

Furthermore, we complement the portfolio sort with a panel regression:

$$r_{i,t+1} = \alpha + \beta_1 |\varepsilon_{i,t}| + \beta_2 \hat{r}_{i,t+1} + u_{i,t+1}, \tag{6}$$

with standard errors clustered by month to account for cross-sectional dependence. A significant β_1 would indicate that reconstruction error adds predictive power beyond the model's own fitted return.

Finally, we identify stocks that transition from the lowest anomaly quintile at t to the highest quintile at $t + 1$ ("regime transitions"), and compare their cumulative returns over a 6-month forward window against a control group of stocks that remain in the lowest quintile in both periods.

2.6 Ablation: Energy and Disentanglement Regularization

We propose two auxiliary loss terms. The first penalizes factor Sharpe ratios exceeding a threshold $\kappa = 2$, motivated by no-arbitrage bounds, while he second penalizes off-diagonal elements of the factor correlation matrix to encourage orthogonal factors:

$$\mathcal{L}_{\text{energy}} = \lambda \max\left(0, \text{SR}^2 - \kappa^2\right), \quad \mathcal{L}_{\text{disentangle}} = \gamma \left\| \text{Corr}(\hat{\mathbf{F}}) - \mathbf{I} \right\|_F^2. \tag{7}$$

We conduct a grid search over $\lambda \in \{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$ and $\gamma \in \{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$, training a single-seed CA_2 with $K = 5$ for each cell.

3. Results

3.1 Model Comparison

Figure 2 reports out-of-sample performance across all 30 architecture–factor configurations. The key findings mirror and extend Gu et al. (2021).

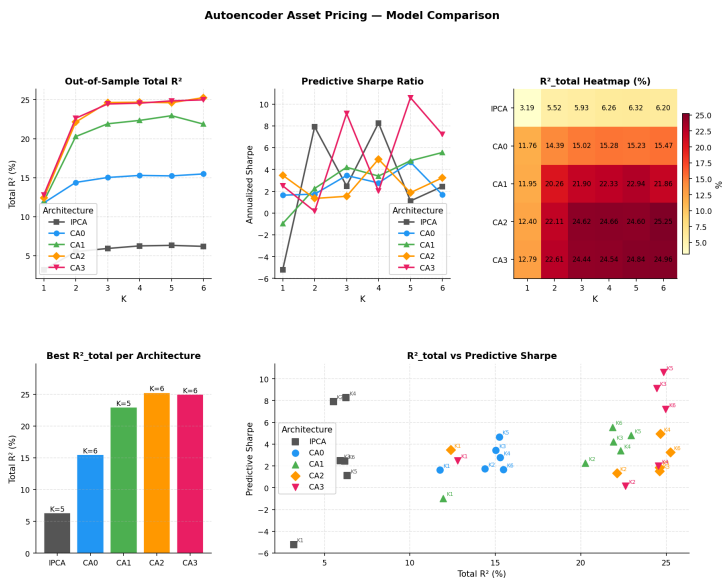

Figure 2: Out-of-sample performance across architectures and latent dimensions. Top-left: total R^2 increases with depth and saturates at CA_2/CA_3. Top-center: predictive Sharpe ratios are noisy, with CA_3 showing the widest variance. Bottom-left: best R^2 per architecture. Bottom-right: R^2 vs. predictive Sharpe shows weak correlation, indicating that better in-sample fit does not reliably translate to higher tradable performance.

Total R^2 increases with depth and saturates. Moving from IPCA (max 6.3%) to CA_0 (15.5%) to CA_1 (23.0%) to CA_2/CA_3 (25.0–25.4%) shows large gains from the first two hidden layers, with diminishing returns thereafter. The nonlinear beta specification captures roughly four times more return variation than the linear model, consistent with the original paper's findings.

Predictive R^2 is modest across all architectures. The best predictive R^2 values cluster around 0.5–2.8%, reflecting the well-known difficulty of predicting individual stock returns. The gap between total and predictive R^2 (e.g., 25.1% vs. 0.65% for CA_2 $K = 5$) indicates that most explained variation comes from contemporaneous factor realizations rather than lagged conditional betas.

Predictive Sharpe ratios are noisy across configurations. The median predictive Sharpe across all 30 configurations is 2.63, with an interquartile range of [1.69, 4.77]. Twenty-six of 30 configurations exceed the characteristics-only benchmark of 1.24, demonstrating that the autoencoder consistently adds value beyond a naive characteristic composite. However, the large variance across K and architecture (particularly for CA_3, where the Sharpe ranges from 0.16 ($K = 2$) to 10.59 ($K = 5$)) suggests that deeper networks are more sensitive to the choice of latent dimension. The median is more representative than the maximum, as a full out-of-sample test would require pre-committing to an architecture.

3.2 Factor Interpretation

Table 1 reports correlations between the estimated latent factors and macroeconomic variables. Both the linear and deep models anchor primarily on the unemployment rate and the 10-year Treasury yield, with absolute correlations ranging from 0.1 to 0.3. This is economically intuitive, as the business cycle (proxied by unemployment) and the

Table 1: Top 3 Macroeconomic Drivers by Factor

Factor	Linear Model (CA_0)	Deep Model (CA_2)
1	UNRATE (0.30), GS10 (−0.25), FEDFUNDS (−0.15)	UNRATE (−0.24), GS10 (0.10), DEXUSEU (−0.07)
2	UNRATE (−0.26), GS10 (0.23), FEDFUNDS (0.09)	GS10 (−0.16), UNRATE (0.11), VIXCLS (0.09)
3	UNRATE (−0.22), GS10 (0.10), CPIAUCSL (0.04)	UNRATE (−0.20), GS10 (0.10), VIXCLS (0.02)
4	GS10 (0.26), UNRATE (−0.23), FEDFUNDS (0.13)	GS10 (−0.27), UNRATE (0.25), FEDFUNDS (−0.24)
5	UNRATE (0.18), GS10 (−0.09), DGS1MO (0.03)	UNRATE (−0.24), GS10 (0.12), FEDFUNDS (0.05)

Note: Pearson correlations between estimated latent factors and macroeconomic indicators. Factors are estimated from the test period (2011–2023). Macroeconomic data from FRED.

term structure (proxied by long-term yields) are first-order drivers of equity risk premia.

The deep model constructs more specialized factor profiles. Factor 4 in CA_2 loads heavily on the federal funds rate (-0.24), capturing monetary policy exposure that is diffuse across factors in the linear model. Factors 2 and 3 incorporate market volatility (VIX correlations of 0.09 and 0.02), though these modest correlations understate the dynamic relationship.

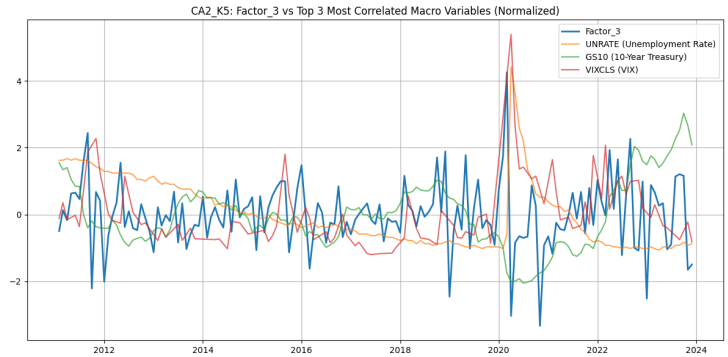

Figure 3: CA_2 Factor 3 time series versus the VIX and unemployment rate. Despite a low unconditional correlation of 0.02 with the VIX, Factor 3 synchronizes with tail events, most notably the COVID-19 shock of March 2020, suggesting it functions as a dynamic tail-risk portfolio.

Figure 3 illustrates this point. CA_2's Factor 3 exhibits sharp spikes coincident with the COVID-19 market crash, tracking both the VIX and unemployment surges despite its near-zero unconditional correlation with the VIX. The autoencoder has learned to construct a tail-risk portfolio from firm characteristics alone, without any macroeconomic variables as inputs.

3.3 Anomaly Detection

Table 2 reports average monthly returns and annualized Sharpe ratios by anomaly quintile. Returns decline monotonically from the lowest-anomaly quintile (0.72%/month, Sharpe 0.64) to the highest (0.36%/month, Sharpe 0.18), producing a high-minus-low spread of −36 basis points per month.

Table 2: Portfolio Sorts by Anomaly Score

Quintile	Mean Return (%/mo)	Std Dev	Sharpe (ann.)	N months
Low anomaly (Q1)	0.717	3.90	0.637	155
Q2	0.630	4.03	0.541	155
Q3	0.626	4.47	0.486	155
Q4	0.529	5.30	0.346	155
High anomaly (Q5)	0.361	6.86	0.182	155
High − Low	−0.356	3.75	−0.328	155

Note: Stocks sorted into quintiles by $|\varepsilon_{i,t}|$ each month; returns measured at $t+1$. Stocks tracked by CRSP permno across months. Model: CA_2, $K = 5$, 10-seed ensemble.

However, the spread is not statistically significant. The Newey-West t-statistic is -1.07 ($p = 0.28$), the standard t-test gives $t = -1.18$ ($p = 0.24$), and the Wilcoxon signed-rank test yields $p = 0.069$. The monotonic pattern is suggestive but lacks the statistical power to reject the null of zero spread, likely due to the high volatility of the anomaly quintiles.

The panel regression of $r_{i,t+1}$ on the absolute residual and the model's fitted return yields a significant negative coefficient on the anomaly score ($\hat{\beta}_1 = -0.0061$, $t = -5.23$ with month-clustered standard errors, 155 clusters). The fitted return passes through nearly one-for-one ($\hat{\beta}_2 = 1.018$, $t = 68.5$), and the regression R^2 of 25.1% reflects the model's strong contemporaneous fit.

The negative sign on the anomaly coefficient indicates that high-anomaly stocks earn lower future returns, controlling for the model's prediction. This is consistent with the portfolio sort finding and suggests that large reconstruction errors capture uncompensated idiosyncratic risk or noise, not exploitable mispricing. In other words, the autoencoder's residuals are economically informative (they correlate with future underperformance) but not profitably tradable after accounting for the model's own predicted return.

Of the 22,660 stock-months in the lowest anomaly quintile, 300 (1.3%) transition to the highest quintile in the following month. These "regime transition" stocks earn a mean cumulative return of 5.97% over the subsequent 6 months, compared to 2.55% for the 22,360 stocks that remain in the lowest quintile. The spread of 1.9 percentage points is directionally positive but statistically indistinguishable from zero (annualized Sharpe of 0.08, standard deviation of 32.5%). With only 300 transitions, the test lacks power to detect a signal of this magnitude.

3.4 Backtest

We backtest the predictive long-short strategy using CA_2 with $K = 5$ over the test period 2011–2023. Each month, we sort stocks into deciles by predicted return $\hat{\boldsymbol{\beta}}_i \cdot \hat{\boldsymbol{\lambda}}_{t-1}$, where $\hat{\boldsymbol{\lambda}}_{t-1}$ is the rolling 60-month mean of

estimated factors seeded with 408 months of training-period factor history.

Table 3: Backtest Performance Summary

Strategy	Ann. Return	Ann. Vol	Sharpe	Sortino	Max DD	Calmar	Hit Rate
Long-Short (AE)	67.9%	31.9%	2.13	3.39	−25.3%	2.69	73.1%
Chars-Only L/S	17.8%	14.3%	1.24	1.14	−32.3%	0.55	70.5%
Long Only	71.6%	36.3%	1.97	2.60	−30.2%	2.37	76.3%
Short Only	3.6%	7.6%	0.48	0.36	−19.6%	0.19	68.6%
Market (EW)	6.9%	16.7%	0.41	0.40	−34.7%	0.20	58.3%

Note: Performance computed over the test period (2011–2023, 156 months). The characteristics-only strategy sorts on the cross-sectional mean of all 91 rank-normalized characteristics without any autoencoder. All portfolios are equal-weighted.

The autoencoder long-short strategy achieves an annualized return of 67.9% with a Sharpe ratio of 2.13 and a 73.1% hit rate. This substantially exceeds the characteristics-only benchmark (Sharpe 1.24), indicating that the autoencoder's nonlinear beta network extracts a characteristic weighting worth approximately 0.89 additional Sharpe ratio units beyond a naive composite. The long leg drives most of the performance (71.6% annually), while the short leg contributes modestly (+3.6% annually); the short side of the portfolio selects stocks with the lowest predicted returns rather than stocks predicted to decline in absolute terms.

The CAPM alpha is 47.5 basis points per month ($t = 7.76, p < 10^{-14}$), with a market beta of $+1.58$ and an R^2 of 0.68. The positive beta reflects the strategy's net long bias: the long leg has substantially higher volatility (36.3%) than the short leg (7.6%), and the returns are positively skewed (0.98), consistent with a momentum-driven stock selection strategy in a period dominated by growth outperformance.

Mean monthly turnover is 29.7% for the long leg and 30.2% for the short leg, corresponding to an average holding period of approximately 3.3 months. The Jaccard similarity between consecutive months' holdings is 0.54, indicating that roughly half of the portfolio persists month-to-month. While the portfolio does rotate, a persistent core of holdings remains stable: the most frequently longed stock appears in 137 of 156 months (88%), and the most frequently shorted stock appears in 153 of 156 months (98%). The strategy is thus primarily a cross-sectional stock selector, rather than a high-frequency factor timer.

To guard against model selection bias, we report the median predictive Sharpe across all 30 architecture–factor configurations: 2.63 (IQR [1.69, 4.77]). Twenty-six of 30 configurations exceed the characteristics-only benchmark. The variance across configurations, particularly the CA_3 range of [0.16, 10.59], motivates caution in interpreting any single model's Sharpe.

3.5 Ablation Studies

Table 4 reports total R^2, predictive R^2, and validation loss across the 5×5 grid of energy penalty (λ) and disentanglement penalty (γ). The results are based on single-seed CA_2 with $K = 5$.

The best-performing cell ($\lambda = 0.1$, $\gamma = 0.1$) achieves the highest total R^2 (25.3%) and predictive R^2 (12.7%) in the grid, improving over the unregularized baseline (22.7% and 10.6%). Several cells with moderate regularization also outperform the baseline—for instance, $\lambda = 0.001$, $\gamma = 0.01$ achieves 25.1% total R^2—

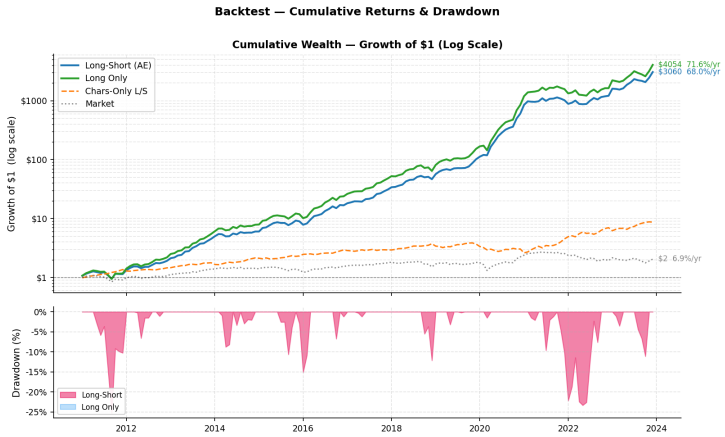

Figure 4: Top: cumulative returns for the long-short, long-only, and equal-weighted market strategies. The exponential growth reflects compounding of 67.9% annualized returns. Bottom: drawdown profile showing the maximum drawdown of −25.3% in September 2011, with subsequent drawdowns during 2015–2016, 2019–2020, and 2022.

Table 4: Ablation: Energy and Disentanglement Regularization (CA_2, $K = 5$)

λ	γ	R^2_{total} (%)	R^2_{pred} (%)	Val. Loss
0	0	22.7	10.6	0.0206
0	0.01	24.1	11.4	0.0207
0.001	0.01	25.1	12.1	0.0209
0.0001	0.1	23.8	11.6	0.0207
0.1	**0.1**	**25.3**	**12.7**	**0.0207**
0.01	0.1	21.0	8.7	0.0213

Note: Selected cells from the full 5×5 grid. Best configuration in bold. Single random seed per cell.

suggesting that the energy and disentanglement penalties can help generalization. However, over-regularization can be harmful: $\lambda = 0.01, \gamma = 0.1$ degrades total R^2 to 21.0%.

The ablation results are inherently noisy with single-seed training. The best cell in a prior run with different random initialization was ($\lambda = 0.01, \gamma = 0.01$), and the two runs' grids show no consistent monotone pattern. This instability underscores the need for multi-seed experiments to draw firm conclusions about optimal regularization strength.

4. Discussion

Our replication confirms the core finding of Gu et al. (2021): conditional autoencoders with nonlinear beta specifications substantially outperform linear factor models in explaining cross-sectional return variation. The total R^2 advantage of deep models (25% vs. 6% for IPCA) is large and robust across K, consistent with the idea that characteristic interactions and nonlinearities are economically important for mapping firm attributes to risk exposures.

The decomposition of the predictive Sharpe ratio into autoencoder and characteristics-only components provides a cleaner picture of the model's economic value. The characteristics-only benchmark of 1.24 represents the Sharpe ratio available from sorting on a simple average of rank-normalized characteristic, a strategy that requires no machine learning. The autoencoder's predictive Sharpe of 2.13 (median across configurations: 2.63) exceeds this benchmark, indicating that the nonlinear beta network captures interactions among characteristics that a simple composite misses. The turnover analysis reveals a strategy that is predominantly a cross-sectional stock selector with moderate rebalancing: mean monthly turnover of 30% corresponds to an average holding period of 3.3 months, with a persistent core of holdings that remain in the portfolio for years. The autoencoder has learned a superior but largely stable characteristic weighting, rather than a dynamic factor-timing signal exploiting month-to-month variation in risk premia.

The reconstruction error investigation yields a coherent null result. While anomaly scores produce a monotonic return gradient across quintiles, the high-minus-low spread is not statistically significant. The negative predictive regression coefficient indicates that high-anomaly stocks underperform, but this effect is subsumed by the model's own predicted return. This is consistent with the autoencoder being a well-specified model: its residuals are close to white noise, leaving little exploitable signal for a secondary trading strategy. The transition analysis corroborates this interpretation; stocks that abruptly shift from low to high anomaly earn directionally higher returns, but the spread is not significant with only 300 observed transitions.

Several limitations deserve mention. First, we use 91 characteristics compared to the 94 in Gu et al. (2021), which may account for small differences in R^2 levels. Second, our backtest does not account for transaction costs; with 30% monthly turnover and an average holding period of 3.3 months, costs would reduce the net Sharpe ratio but are unlikely to eliminate the spread over the characteristics-only benchmark. Third, the large variance in predictive Sharpe across configurations (IQR of 3+ units) highlights sensitivity to model selection that would attenuate in a genuine real-time implementation where the architecture must be pre-specified. Fourth, the single-seed ablation study provides only suggestive evidence on the value of energy and disentanglement regularization, as the best cell varies across random initializations.

Future work could extend this framework in several directions: incorporating options data to compute the variance risk premium for regime-transition stocks, implementing value-weighted portfolios to assess capacity constraints, running multi-seed ablation studies to establish the statistical significance of the regularization improvements, and decomposing the predictive Sharpe into stock-selection and factor-timing components to quantify the autoencoder's incremental contribution along each dimension.

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